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Exercise – *Queen of Hearts*

"My dear, here we must run as fast as we can, just to stay in place. And if you wish to go anywhere you must run twice as fast as that."

Alice's Adventures in Wonderland *by Lewis Carroll (1832–1898)*

The Queen of Hearts has been chasing Alice for quite some time, but now she is tired. Alice passes through a gate into a large fenced garden filled with trees and walking pathways. She hopes to tire out her pursuer in the garden so that she can safely escape.

The Queen, however, is determined to catch Alice and employs the rabbits in the garden to help her. Each rabbit r_i has its own, private underground burrow network with n_i nodes and m_i one-way tunnels connecting these nodes. The *home* of r_i is located at node 0 of its network, and c_i carrots are stored there. The network also has an *opening* into the garden, which is located at node $n_i - 1$. Every rabbit is willing to carry its carrots from its home to the opening of its network and deposit the carrots there. Unfortunately, the rabbits are very clumsy: For every tunnel they travel through, they drop one carrot. The queen then can eat the carrots deposited at the openings to regain her energy as she runs through the garden.

The openings of the burrow networks, as well as the entrance and the exit of the garden, are connected by one-way walking pathways. You can assume that initially, the queen has enough energy left to run along one single pathway starting at the entrance gate of the garden. From there on, whenever the queen wants to continue the chase and run along another pathway, she must first eat one carrot to gain enough strength. She can only eat one carrot at a time and she cannot carry any carrots with her.

Alice notices the carrots deposited at the openings and how they might foil her plans. Thus, she decides to steal some of them so as to ensure that the queen cannot reach the exit gate of the garden. On a normal day, the rabbits do not care and let Alice steal the carrots. But on some days they are grumpy, and then Alice has to pay one gold coin to each rabbit whose carrots she steals. Obviously, Alice wants to steal as few carrots or pay as little gold as possible.

Input The first line of the input contains the number $t \leqslant 30$ of test cases. Each of the t test cases is described as follows:

- It starts with a single line containing three integers `r m d`, separated by a space, where r, for $1 \leqslant r \leqslant 200$, denotes the number of rabbits (and burrow networks), m, for $2 \leqslant m \leqslant 10^4$, denotes the number of walking pathways in the garden, and d indicates whether it is a normal day ($d = 0$) or the rabbits are grumpy ($d = 1$).
- The following r blocks of lines each define a rabbit's burrow network:
  - The first line of each block contains three integers n_i m_i c_i, where n_i and m_i denote the number of nodes and tunnels in rabbit r_i's network, respectively, and c_i denotes the number of carrots rabbit r_i has at his home ($1 \leqslant n_i \leqslant 200$, $1 \leqslant m_i \leqslant 1600$, $0 \leqslant c_i \leqslant 300$).

- The next m_i lines define the tunnels of r_i's network. Each line contains two integers u v, separated by a space, that define a one-way tunnel from u to v in the respective network ($0 \leqslant u, v < n_i$).

- The following m lines define the walking pathways of the garden. Each line contains two integers u v, separated by a space, that define a one-way walking pathway from u to v in the garden. Integer 0 denotes the entrance gate of the garden, integer $r+1$ refers to the exit gate of the garden, and any integer i, where $1 \leqslant i \leqslant r$, refers to the opening of rabbit r_i's network.

Output The output for each test case consists of a separate line containing two integers c g. The first integer c denotes the minimum number of carrots that Alice has to steal (regardless of the number of gold coins she has to pay to the rabbits) such that the Queen cannot reach the exit gate of the garden. The second integer g denotes the minimum number of gold coins Alice has to pay to the rabbits such that she can steal sufficiently many carrots (possibly more than c) so as to ensure that the Queen cannot reach the exit gate of the garden. Note that $d = g = 0$ on a normal day.

Points There are four groups of test sets worth 80 points in total. For each group there is also a corresponding hidden test set that is worth 5 points. So, you can achieve $80 + 4 \cdot 5 = 100$ points overall.

1. For the first group of test sets, worth 20 points, you may assume that $r = 1$, i.e., there is exactly one rabbit in the garden, and there is a walking pathway from the entrance of the garden to the opening of this rabbit's network, and another walking pathway from this opening to the exit gate of the garden. In addition, you may assume $d = 0$.

2. For the second group of test sets, worth 20 points, you may assume that every opening of a burrow has at most one walking pathway leading into it and at most one walking pathway leading out of it. In addition, you may assume $d = 0$.

3. For the third group of test sets, worth 25 points, you may assume $d = 0$.

4. For the forth group of test sets, worth 15 points, there are no additional assumptions.

Sample Input

```
1
3 6 1
3 2 5
0 1
1 2
2 1 2
0 1
4 4 10
0 1
1 2
2 3
3 0
0 1
0 2
1 2
1 3
2 3
3 4
```

Sample Output

```
4 1
```